Entangled Minds: Case Report

The Quantathon · Qubits Qrew · NED University
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1. Final Theory / Verdict

My conclusion, with approximately 95% confidence, is that ECHO did not send a usable message faster than light. The most likely explanation is that Dr. Zara Mahmud misunderstood quantum correlations as proof of faster-than-light communication. Quantum entanglement can create strong connections between measurement results, but the results at each location are random on their own. Therefore, entanglement alone cannot be used to send a message faster than light. Normal communication through a classical channel is still required. This conclusion is based on the physics evidence and does not identify anyone as responsible for any human action.

2. Evidence and Clues

- The Module 03 entangled-pair measurements produced only 00 and 11 in the same measurement basis, showing strongcorrelations between the two qubits.
- The CHSH score was 2.8194, above the classical bound of 2 and below the quantum maximum of approximately 2.8284.This is consistent with a quantum violation of the classical CHSH limit.
- In Module 05, majority voting recovered the bit string 01000101010000110100100001001111, which decodes to ECHO.This demonstrates recovery of a classically repeated message, not faster-than-light transmission.
- In Module 06, teleportation with corrections gave 100% receiver accuracy. Without corrections, the measured accuracywas 51%, consistent with the expected 50% chance baseline for 2,000 shots.

3. Evidence Board

Category	Finding
FACT	The simulator returned the measurement counts and scores recorded in the notebook: Module 03 CHSH = 2.8194; Module 05 decoded ECHO; Module 06 returned 100% with corrections and 51% without.
CLAIM	The alleged ECHO event involved a message sent faster than light. The simulation results alone do not establish that such an event occurred.
INFERENCE	The results support quantum correlations and the need for classical correction information in teleportation. They do not support usable faster-than-light communication.
OPEN QUESTION	What exactly did Dr. Mahmud observe in the original event, and were the real experimental records, timing, and classical communication channels independently verified?

4. Experiments and Results

Module 03 — Entanglement and CHSH: A Bell pair was prepared with a Hadamard gate and CNOT, then measured. Bronze counts were 00: 983 and 11: 1017; Silver counts were 00: 1005 and 11: 995. The CHSH test gave S = 2.8194, compared with a classical limit of 2 and quantum maximum of $2\sqrt{2} \approx 2.8284$.

Module 05 — Damaged message: The 96-bit stream was divided into 32 triples. Majority voting reconstructed 01000101010000110100100001001111, which decodes as ECHO. A triple such as 110 demonstrates how two corrupted copies can reverse a majority decision when the original bit was 0.

Module 06 — Teleportation: The receiver's test gave 100% accuracy with classical corrections and 51% without them. The small deviation from 50% is expected from finite sampling. The comparison shows that entanglement alone does not provide the receiver with a usable copy of the intended state.

5. Interpretation and Alternative Explanation

The combined evidence fits standard quantum theory: entanglement creates nonclassical correlations, while teleportation requires classical measurement information to complete the transfer. A plausible alternative is that the original observation was caused by an ordinary timing error, an unaccounted classical channel, or a mistaken interpretation of correlated outcomes. The available notebook simulations cannot decide which, if any, of these occurred in the real incident. I prefer the no-faster-than-light-signalling explanation because it matches the tested circuits and established quantum principles.

6. Next Experiment

Repeat the Bell test with documented measurement settings, many independent runs, and a pre-registered CHSH calculation. Separately, audit the alleged message experiment by recording detector timestamps and all classical communication paths. The key test is whether the receiver's local outcomes alone depend on the sender's chosen message before any classical information can arrive. Quantum theory predicts that they do not.

7. Limitations

These are simulator results, not a reproduction of the alleged real-world event. Finite shots introduce statistical variation. A CHSH violation supports nonclassical correlations but does not demonstrate faster-than-light signalling or identify a cause, motive, or responsible person. The investigation therefore supports a physics conclusion, not a definitive account of every detail of the incident.

8. References and AI / External Assistance

- The Quantathon, Qubits Qrew, NED University, *ECHO Participant Manual* and *ECHO Investigation Notebook* supplied for the challenge.
- Qiskit documentation for quantum circuits, measurement, and simulation: https://docs.quantum.ibm.com/
- AI assistance: ChatGPT (OpenAI) was used to help explain quantum concepts, debug and interpret notebook outputs,and organize and edit this report. The participant should verify all results and ensure this disclosure complies with event rules.